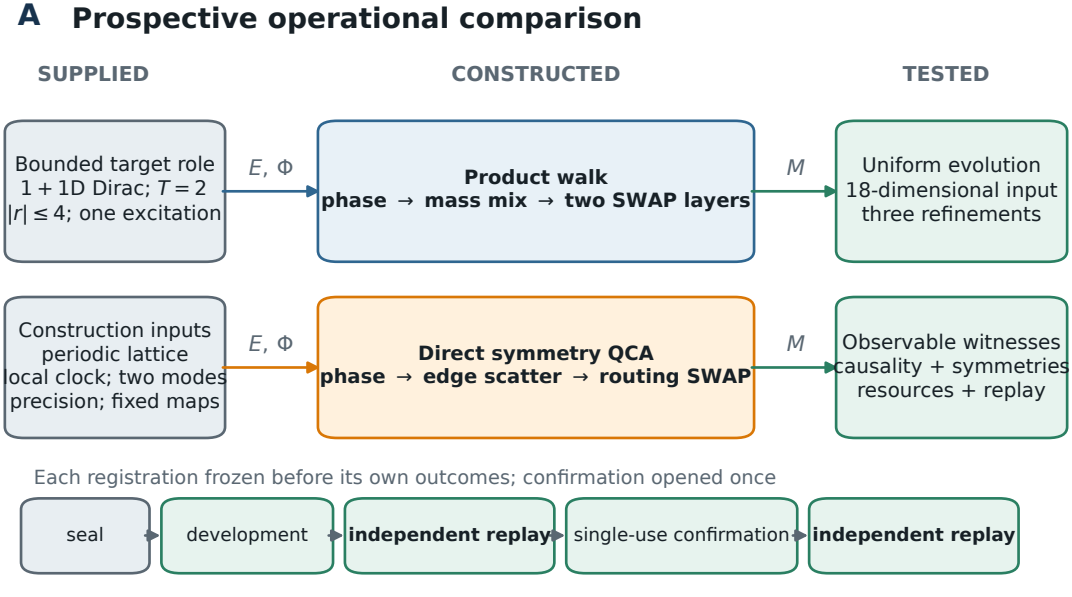

A
Prospective operational comparison
SUPPLIED
CONSTRUCTED
TESTED
Bounded target role
1 + 1D Dirac; T = 2
|r| ≤ 4; one excitation
E, Φ
Product walk
phase → mass mix → two SWAP layers
M
Uniform evolution
18-dimensional input
three refinements
Construction inputs
periodic lattice
ocal clock; two modes
precision; fixed maps
E, Φ
Direct symmetry QCA
phase → edge scatter → routing SWAP
M
Observable witnesses
causality + symmetries
resources + replay
Each registration frozen before its own outcomes; confirmation opened once
seal
development
independent replay
single-use confirmation
independent replay